

TFPT — Research Contracts for the two open gates

(U_{wall}) : the parabolic flavor wall-selection and (G_{metric}) : the full quantum-gravity measure

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What this document is about

After the compiler closure the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. (F_{frontier}) (Koide, η_B , axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** (U_{wall}) *first* (finite, algebraic, falsifiable), then (G_{metric}) (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), U_{point} collapses to the *one* dimensionful unit v_{geo} shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where U_{point} still appears); $(G_{\text{metric}}) \rightarrow G_{\text{net}}$: holographically reduced to the single boundary-net statement “*the seam-Calderón inclusion has Jones index 4 = $|\mu_4|$* ” — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/GATE.METRIC.05-07); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank = c ; Wick/Pfaffian exact) is the certified Calderón datum — closing G_{net} therefore also closes the carrier choice [L] ([`verification/v113_quasifree_kernel.py`]); $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$: named QFT/cosmology *transfer* interfaces, never compiler powers.

Research Contract 1 — (U_{wall})

Goal. Prove

$$\nabla_F^* = \text{Sel}_{D_4, \text{cusp}, \det R=8, \text{Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4 . The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4 -symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation). *The gate data are equivalent to the $SU(3)$ character variety $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C_{\text{cusp}}^4 : M_1 M_2 M_3 M_4 = \mathbf{1}\} / SU(3)$ with $C_{\text{cusp}} = \text{Ad}_{SU(3)} \text{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, and $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1 \rangle$. To show: $\mathcal{M}_{\text{par}}^{D_4} \simeq \mathcal{X}_{\text{cusp}}^{D_4}$ as polystable*

Mehta–Seshadri / tame Hodge realisation. Machine: the group presentation and D_4 -action are finite and codable (Sage/Wolfram).

Lemma 2 (U1 — wall landscape, corrected). *Parabolic degree zero forces the line weight-sums onto the wall $(w_1, w_2, w_3) = (2, 1, 1)$. An explicit balanced representative (weights in $\frac{1}{3}$) is*

$$W_{\text{wall}} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

*each column a permutation of $(0, 1, 2)$, row sums $(2, 1, 1)$. **Honest result:** the $6^4=1296$ column-permutation matrices give 144 walls; under the full group $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$ they form five orbits, not one. So W_{wall} is not singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set; W_{wall} is one of the five. Machine: fully certified $C_U^{(1)}$ ([verification/v29_research_contract_certs.py]).*

The splitting type is now algebraic (RH route, [verification/v32_rh_splitting.py]). *Passing from the monodromy to the Fuchsian residues A_k (eigenvalues = weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, $A_k = U^{k-1}A_0U^{-(k-1)}$, $U = \text{diag}(1, i, -i)$) the twisted $\mathbb{Z}_4$ -average collapses exactly: $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$, so the exponents at infinity are $4 \text{diag}(A_0)$. A flat bundle with regular ∞ needs integer exponents summing to $4 = -\deg E$; with the cusp weights the only options are perms of $(2, 1, 1)$ and $(2, 2, 0)$, i.e.*

$$\boxed{\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \text{diag}(A_0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})}$$

(the sibling $(2, 2, 0) \rightarrow \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$). Such a Hermitian A_0 (spectrum $\{0, \frac{1}{3}, \frac{2}{3}\}$, that diagonal) exists by Schur–Horn. So the splitting is an algebraic condition on $\text{diag}(A_0)$. Honest wall: the global $\prod M_k = \mathbf{1}$ is the path-ordered monodromy (not $\exp(2\pi i \sum A_k)$), constraining off-diag(A_0) via the genuine RH solve; and R-extraction still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ bridge. The RH route fixes the splitting algebraically but not c_u/c_d .

Lemma 3 (U2 — the kill switch, run and hardened [N]). *The selected polystable wall representative must not collapse to the diagonal abelian $U(1)^3$ point if it is to give non-trivial $SU(3)_F$ transport mixing: (A) ∇_F^* polystable but not simultaneously diagonal (the desired breakthrough), or (B) diagonal collapse, in which case c_u/c_d stays honestly [P]. **Result: the cheap test is run robustly and lands on A.** The kill-switch fires iff the D_4 -symmetric cusp representation is reducible (commutant non-scalar). Across 400 random cusp-class generators of the $\mathbb{Z}_4$ family $M_k = U^{k-1}MU^{-(k-1)}$ the commutant is scalar at every point (irreducible 400/400), the reducible locus is hit with probability 0 (measure-zero, codimension ≥ 1), and the mixing strength stays bounded away from zero (min off-diag = 0.35 > 0.05). So case A is the generic behaviour, not an accident of the one explicit point of [verification/v33_explicit_flat_bundle.py]: (U_{wall}) survives its own cheapest falsification test. [verification/v38_uwall_killswitch.py]*

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, positive-dimensional). *With $M_k = U^{k-1}AU^{-(k-1)}$ ($U = \text{diag}(1, i, -i)$, the $\mathbb{Z}_4$ rotation) and the reflection $M_1 = VM_1^{-1}V^{-1}$, $M_3 = VM_3^{-1}V^{-1}$, $M_2 = VM_4^{-1}V^{-1}$ (with $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$, $V^2=1$, $VUV^{-1}=U^{-1}$), the relations become polynomials in the entries of $A \in C_{\text{cusp}}$; $\mathcal{X}_{\text{cusp}}^{D_4}$ is cut out as $\bigcup_k C_k$, parametrised by trace coordinates $x = \text{tr}(M_1 M_2)$ (note $\text{tr } A = 1 + \omega + \omega^2 = 0$). **Result** ($C_U^{(2)}$, [verification/v30_d4_character_variety.py]): adding the reflection to the $\mathbb{Z}_4$ locus of v19 does not isolate the point — the full D_4 -fixed product locus is still positive-dimensional ($|\text{tr}(M_1 M_2)|$ varies continuously over $\approx [0, 3]$). So even the full D_4 symmetry cannot select ∇_F^* . Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).*

Lemma 5 (U4 — selectors as equations). *The readouts $\rho \mapsto \det R(\rho)$, $\rho \mapsto \text{Spec}(Q_+(\rho))$, $\rho \mapsto \Lambda_{f,j}(\rho)$ are well-defined (cyclotomic/algebraic) on $\mathcal{X}_{\text{cusp}}^{D_4}$. **Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4} : \det R = 8, \text{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not closed. **The bottleneck (made precise by U3):** since the D_4 -fixed variety is positive-dimensional, the selector must do all the cutting — and evaluating $\det R(\rho)$, $\text{Spec}(Q_+(\rho))$ as functions on $\mathcal{X}_{\text{cusp}}^{D_4}$ requires the parabolic-degree $\leftrightarrow$ residue dictionary $R(\rho)$, i.e. the H2 equivalence. That dictionary is the single remaining analytic input of the (U_{wall}) gate.

What $R(\rho)$ is ([verification/v31_R_dictionary.py]). Two facts pin its nature. (i) Case A is alive: every D_4 -fixed cusp monodromy found is irreducible, so the wall point carries non-trivial $SU(3)_F$ mixing — the U2 kill switch does not fire to the diagonal case B. (ii) $R(\rho)$ is not algebraic: R is integer-valued, but the algebraic invariants of ρ (e.g. $\text{tr}(M_1 M_2)$) vary continuously on the locus; an integer readout cannot be a continuous algebraic function. Hence $R(\rho)$ is the discrete invariant of the non-abelian Hodge / Mehta–Seshadri map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle E_ρ): integer output, but its computation is the transcendental harmonic-metric (Hitchin) solve on the generic Higgs locus. **Refinement (U_4'' ,** [verification/v40_harmonic_metric.py]): at the polystable point TFPT actually selects this pessimism does not apply — see the box below.

Progress: the selector side of U4 is closed on the explicit point [N]

While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 selectors are *algebraically* accessible on the explicit flat bundle of [verification/v33_explicit_flat_bundle.py] — no PDE. Computed exactly from A_0 : the exponents at infinity $\text{eig}(\sum_k A_k) = \{2, 1, 1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic *anchor* $a = (1, 1, 2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\begin{aligned} \det R &= 8 = n \cdot a \quad (\text{anchor pairing on the splitting type}), \\ \text{Spec}(Q_+) &= \{1, 2, 3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{aligned}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\text{Spec}(Q_+)$ is the affine image ($d=3$ the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data*. **Residual (sharpened below by Readout Rigidity):** the quark ratio c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [verification/v39_uwall_selectors.py]

Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE [N]

The U4 “transcendental Hitchin solve” is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33_explicit_flat_bundle.py] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim \ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So $C_U^{(3)}$ reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length R (closed: H1 + $\det R = 8$).** *Residual:* the feared PDE is gone; the quark ratio c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40_harmonic_metric.py]

The final leg-assignment test (honest result) [N]

Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1 , so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R); the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41_leg_assignment.py]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem [I]/[N] ([verification/v114_torsion_delta.py])

The v41 “genericity is future work” item is now executed — and yields a theorem. *(i) Flatness* $= \mu_4$ *torsion* [I]: for any M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$ -family flatness constraint *is* the torsion statement “ $T = MU$ is a fourth root of unity”: the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem* [I] (both directions): on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1}v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2}), \quad \text{spec } M = \{1, \omega, \omega^2\} \text{ automatic}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + det 1). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a μ_4 -torsion *identity*, not an observation. *(iii) Reflection lemma* [I]: the D_4 reflection forces $\text{diag } M = (r, z, \bar{z})$ with r real and $\text{Re } z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census* [N] (seeded): the full D_4 -fixed flat cusp locus realises only $\text{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue* [P] (recorded): whether the anchor splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item.

The anchor pins the residue: an exact normal form for the (U_{wall}) bundle [I]/[N] ([verification/v115_anchor_residue.py])

The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i) μ_4 -average lemma* [I]: $\sum_k U^k X U^{-k} = 4 \text{diag } X$ for any X — μ_4 conjugation-averaging *is* the diagonal projection, so the exponents at infinity are $4 \text{diag } A_0$ and the anchor splitting holds *iff* $\text{diag } A_0 = (2, 1, 1)/4 = a/|\mu_4|$: the residue diagonal **is** the anchor over μ_4 (v33’s diagonal was forced, not chosen). *(ii) The $(8, 0, 5)/144$ lemma* [I]: the anchor diagonal and the cusp spectrum $\{0, 1, 2\}/N_{\text{fam}}$ pin $\sum |a_{ij}|^2 = \frac{13}{144}$, and with $a_{13} = 0$

the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2, |a_{13}|^2, |a_{23}|^2) = \frac{(8, 0, 5)}{144}} \quad \begin{array}{l} 8=\text{rank } E_8, \ 5=g_{\text{car}}, \\ 144=(|\mu_4|N_{\text{fam}})^2, \ 8+5=13=\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$. (iii) *Exact normal form [I]*: $A_0^* = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$ has $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ *exactly*. (iv) A_0^* is the RH solution [N]: its big-circle monodromy is trivial to 10^{-10} , $\prod M_k = \mathbf{1}$, and the unitarised holonomy has $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, $\text{tr}(\widetilde{M}_0 U) = 1$ — the v33/v40 point is gauge-equivalent to A_0^* (the a_{12}, a_{23} phases are diagonal gauge). (v) *The anchor forces the $d_1 = 0$ slice [N]*: a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**; $\delta = \frac{1}{2}$ needs no selection. *Residue [P]*: $a_{13} = 0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [I]/[L] via Gröbner on the RH constraints is the remaining U_{point} step.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar [U]/[P]

The v41 negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r, w) = (1, 1)$ ($L = 7$ for both), and the C_6 resolvent $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos \frac{r\pi}{3})$ reads *only* (r, w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\text{Pl}_{1,a}(K)$. The “11” is then *generated*, not read:

$$(K, \mathbf{1}, a) \rightarrow B_K = \left(\begin{smallmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{smallmatrix} \right) \rightarrow \text{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \rightarrow \|\text{Pl}(K)\|_1 = 11,$$

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}} \quad (\text{anchor microcode} = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}).$$

Typing: [I] for the Plücker / microcode identities ($\text{Pl}(K) = (-1, 6, 4)$, $\|\cdot\|_1 = 11$, $5 \cdot 11/(9 \cdot 13) = 55/117$); and — by Readout Rigidity (v49, below) — [I] also for the *physical* ratio $C_{ud}(\rho^*) = \text{this } \Lambda^2 F \text{ readout}$, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from “find the right scalar y ” to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale (U_{point}) stays as an anchor, no new scalar, no fishing. [verification/v42_exterior_leg.py]

Bridge test ([verification/v43_exterior_bridge.py]). The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \widetilde{M}$ exactly — the exterior leg *is* the $\bar{\mathbf{3}}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$ does *not* equal the *integer* $\text{Pl}(K) = (-1, 6, 4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^* \rightarrow \text{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R ; $\det R = 8$ and $\text{Spec}(Q_+) = \{1, 2, 3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([verification/v44_carrier_exterior.py]). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The “ Λ^2 ” of the exterior leg is

therefore not exotic Hodge data — it is the carrier’s own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from “discrete non-abelian-Hodge invariant” to “the carrier exterior degree” — a standard [L] fact. *Honest scope*: this grounds the *type*; the Plücker *number* 55/117 stays the family $\text{Pl}(K)$ [I], and the carrier- $\Lambda^2 \leftrightarrow$ family- $\Lambda^2 F$ link (via family $\otimes$ carrier = 15 = 16–1) plus the normalisation remain [P].

The “11” is Pascal — a third, simplest origin ([verification/v45_family_exterior.py]). Following that link, $\text{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3 = SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row 4 = (1, 4, 6, 4, 1) with full sum $2^4 = 16 = \dim S^+$, and $|\text{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\|\text{Pl}(K)\|_1 = \sum_{k=0}^2 \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree 2 = $\deg(u^c)$), and family $\otimes$ carrier = 15 = $\dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$ read three ways. So the “11” is generated a *third* way — the cumulative family exterior algebra up to the up-quark’s degree — pure Pascal/branching, *no special math*. The [P] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the “11” worry, *discharged* by Readout Rigidity). *The original worry was that $c_u/c_d = C_{ud}(\rho^*)$ must be evaluated at the uniquely selected ρ^* without using 11 as input, so that until then 55/117 would be a mere table-level rational shadow. **This is now resolved (v49):** C_{ud} is constant on the discrete selector stratum $S_{8,Q}$, so the “11” is earned as the rigid Plücker readout $\|\text{Pl}(K)\|_1$ (v42) — independently of point-uniqueness. Point-uniqueness of ρ^* enters only the absolute amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates). $C_U^{(1)}$ wall enumeration (done); $C_U^{(2)}$ D_4 -fixed character solver; $C_U^{(3)}$ selector uniqueness (one S -equivalence class); $C_U^{(4)}$ amplitude extraction — the ratios $R, Q, c_u/c_d$ are read off the (derived) selector stratum (closed, v49/v71); only the absolute U_f^*, c_q normalisation is the anchor.

Progress: an explicit valid flat bundle is realised (RH solve) [N]

A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, $\text{diag} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)} / (z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \text{diag}(2, 1, 1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1 M_2 M_3 M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid (U_{wall}) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope*: this is one realising point (existence + case A), likely one of a residual family; the unique ∇_F^* still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). [verification/v33_explicit_flat_bundle.py]

H2 bridge probed ([verification/v34_h2_bridge_attempt.py]). The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\text{diag } M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes (0.475, 1.107, 0.917). So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family ρ^*) is the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117.

Theorem U — the four-way split (sharper than one monster gate) [F]/[A]

D_4 fixes the admissible chamber, not the physical point (the D_4 -fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\text{SNF}(R) = (1, 1, 8)$, $\text{Spec}(Q_+) = \{1, 2, 3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} [N]: parabolic degree 0 + polystable $\Rightarrow$ unitary (Mehta–Seshadri) $\Rightarrow \Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([verification/v40_harmonic_metric.py]).
- U_{H_2} [N]: R, Q, K are read off the bundle (splitting type = a , $\det R = 8$, $\text{Spec}(Q_+) = \{1, 2, 3\}$) ([verification/v39_uwall_selectors.py]).
- U_{Λ^2} [I]: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \text{SNF}(R) = (1, 1, 8), \text{Spec}(Q_+) = \{1, 2, 3\}\}$ the anchor-plane readout is *constant*, $C_{ud}(\rho) = g_{\text{car}} \| \text{Pl}_{1,a}(K) \|_1 / (N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([verification/v49_readout_rigidity.py], [verification/v42_exterior_leg.py]).
- U_{point} [A]: full uniqueness of ρ^* — needed only for the *complete* U_f^* amplitude matrix, not for c_u/c_d .

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; only U_{point} stays open.

The simple closure ([verification/v71_simple_r_bridge.py]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \text{rank } E_8$ with $\text{SNF}(R) = (1, 1, 8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\text{fam}}$, v70), and $\text{Spec}(Q_+) = \{1, 2, 3\}$ is the D_4 -equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3, 4, 8, 20)$, product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\text{car}} \cdot 11 / (N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts* — *no transcendental monodromy solve*. The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point} , is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensional scale, not a missing computation.

Gate 1 complete: $U_{\text{point}} \rightarrow v_{\text{geo}}$ ([verification/v75_upoint_to_vgeo.py]). The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c -ratio is closed (Plücker), and each *sector product* is a clean φ_0^{ret} -power (Grand Mass Volume, v46: $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$; the lepton product is $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2g_{\text{car}} / N_{\text{fam}}^2$). Since (pairwise ratios) + (product) $\Leftrightarrow$ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale* v_{geo} . **So U_{point} is not an independent gate: it reduces to v_{geo} , the same one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [A] anchors collapse to one.**

Research Contract 2 — (G_{metric})

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \rightarrow \infty} \int_{\mathcal{G}_\chi / \text{Diff}} \exp(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O) d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \text{Tr} f\left(\frac{D_{\text{rel}} + A_\Sigma}{\chi}\right) - \text{Tr} f\left(\frac{D_{\text{ref}}}{\chi}\right) + \frac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm). *For $s > 5/2$ a Sobolev space of Θ -symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2 T^*M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\text{gh}} = K^*K$ and $\Pi_{\text{phys}} =$*

$\Pi_{\text{BRST/Hodge}}\Pi_{TT}$ is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (not rhetorically). Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

Lemma 9 (G2 — finite relative spectral action, heat-kernel grounded [N]). $S_{\text{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion $\text{Tr } f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \dots$. For the Lichnerowicz Dirac operator $D^2 = \nabla^2 + \frac{R}{4}$ (so $E = -\frac{R}{4}$, spinor trace $\text{tr } \mathbf{1} = 4$) the Gilkey coefficients give, per $(4\pi)^{-2}$,

$$a_2 = \frac{1}{6} \text{tr}(6E + R \mathbf{1}) = -\frac{R}{3} \quad (\text{Einstein–Hilbert } \int R), \quad a_4|_{R^2} = \frac{1}{360} \text{tr}(180E^2 + 60RE + 5R^2 \mathbf{1}) = \frac{R^2}{72}$$

(the R^2 /Starobinsky term), so the spectral action structurally produces $S_{\text{eff}} = \int \sqrt{g} \frac{\bar{M}_{\text{Pl}}^2}{2} (R + \frac{R^2}{6\bar{M}^2}) + \dots$ with the scalaron mass ratio fixed by the cutoff moment, $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$. The TFPT closure $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ fixes $f_0 = 6(4\pi)^2/c_3^7$, i.e. $M = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13} \text{ GeV}$ — the boundary normalisation c_3 is the gravitational scale — and hence the Starobinsky slow-roll forms $n_s \simeq 1 - \frac{2}{N}$, $r \simeq \frac{12}{N^2}$. The $R + R^2$ structure is convention-independent (standard Gilkey); the precise rational $\frac{1}{72}$ and factor 6 are the standard Dirac conventions used here ([verification/v36_spectral_action_g2.py], [verification/v28_gravity_fR.py]).

The simpler resolution: “full QG” for physics is the gap-decoupled admissible sector [F]

The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\text{eff}} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

admissible IR sector = closed under stated RP;
ambient metric measure (G6) = G_{metric} gate

G2 supplies the local action ($R + R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce (G_{metric}) to the ambient projective limit (G6).

Precise status (Alessandro’s boundary): the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure remains a *strict TOE completion target*. Its absence does *not* affect the bounded IR claim, but it *does block certification as a strict physical TOE* — so we do not call it “mere mathematical completeness”. [verification/v36_spectral_action_g2.py]

Lemma 10 (G3–G4 — reflection positivity, fixed and integrated). For each Θ -invariant g in the Calderón ball the physical boundary kernel $\mathcal{K}_g = P_{\text{adm}}\Pi_{\text{phys}}\mathcal{N}_g^{TT}\Pi_{\text{phys}}P_{\text{adm}}$ is reflection positive, $\langle \Theta F, F \rangle_{\mathcal{K}_g} \geq 0$ (G3); and $d\mu_{\text{Cal},\chi}$ is Θ -invariant, supported where G3 holds uniformly, so $\int \langle \Theta F, F \rangle_{\mathcal{K}_g} d\mu_{\text{Cal},\chi} \geq 0$ (G4). **G4 is the decisive step:** RP after integrating over dynamical metrics, not merely at fixed background.

Lemma 11 (G5 — gap dominance [F]). $2\|V_{\text{metric}}\|_{\text{rel}} < \Delta = 6 \log \frac{3}{2}$, with $\|V_{\text{metric}}\|_{\text{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$, hence $2\|V\| = 0.7852 < 2.4328$ and $\Delta_{\text{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$. The numerical inequality is trivial ([verification/v29_research_contract_certs.py]); the work is the operator-norm bound (hard).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full G_{metric}). If the admissible sector is reflection-positive ($G4$) and the metric coupling is gap-dominated, $2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$ ($G5$), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$ protects the admissible sector from the (un-built) ambient/deep-UV, so every physical low-energy read-out (masses, mixings, α^{-1} , the $R+R^2$ regime) is independent of the open ambient measure. Symbolically

$$\boxed{\text{RP} + 2\|V\| < \Delta \implies \text{admissible IR measure stable under ambient completion .}}$$

Thus (G_{metric}) is not a diffuse “full quantum gravity missing” block: it is the sharply delimited ambient projective limit ($G6$), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked [**F**]; the operator-norm bound on $\|V_{\text{metric}}\|$ is the conditional input.

Lemma 12 ($G6$ – $G7$ — projective limit & Ward identities). A projective system μ_{χ_n} with $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$, uniform moment bounds $\sup_n \int \|g\|_{H^s}^p d\mu_{\chi_n} < \infty$, tightness and regulator-independence yields $\mu_{\text{QG}} = \lim \mu_{\chi_n}$ on $\mathcal{G}/\text{Diff}$ ($G6$, the hardest step); the limit obeys the BRST Ward identities $\langle s\mathcal{O} \rangle = 0$, whence $\nabla^\mu T_{\mu\nu} = 0$ and the η boundary phase cancels the APS anomalies ($G7$).

Gate 2 reduction: $G6$ is a *boundary* projective limit, not a bulk one
(`[verification/v76_gmetric_reduction.py]`)

The exact decoupling margin is $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\text{metric}}\|_{\text{rel}} \leq 248 c_3^2 = \dim(E_8) c_3^2 = \frac{31}{8\pi^2} (31 = 2^{g_{\text{car}}} - 1)$. **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so $G6$ is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\text{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\text{metric}} \text{ closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — reduced, not closed. So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) reduced to a boundary measure, no longer a diffuse “full QG missing”. [**P**]/[**A**]

And that boundary measure is a *rigorously constructed net* (`[verification/v77_e8_conformal_net.py]`). The level-1 central charges $c = \dim \mathfrak{g}/(1 + h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank } E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5) + c(A_3) = 5 + 3 = 8 = c(E_8)$ is a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ E_8 -lattice VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the $(E_8)_1$ lattice net* (carrier = $(D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\text{eff}} > 0$ supplies the clustering/tightness — so $G6$ is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [**I**] (charges, embedding) + [**P**] (identification)

Theorem G — the closure statement [**A**] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure μ_{QG} . Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$

seam action

$$f_R R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: open; the deepest item. G5 certified; the regime equation closed [verification/v28_gravity_fR.py].*

Certifiability and recommended order

Building block	Machine	Hardness
W_{wall} enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (<i>done</i>)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy-medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (<i>done</i>)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order

$$(U_{\text{wall}}) \rightarrow \text{freeze the frontier status} \rightarrow (G_{\text{metric}})$$

(U_{wall}) can visibly upgrade the theory — it hits the last concrete flavor ambiguity, including the “11” — and it is finite and falsifiable (the U2 kill switch can destroy it cheaply). (G_{metric}) is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit (G6) the genuine summit.